

BCSE204L	Design and Analysis of Algorithms		L	T	P	C
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Pre-requisite	NIL	Syllabus version				
		1.0				
Course Objectives						
1. To provide mathematical foundations for analyzing the complexity of the algorithms						
2. To impart the knowledge on various design strategies that can help in solving the real world problems effectively						
3. To synthesize efficient algorithms in various engineering design situations						
Course Outcomes						
On completion of this course, student should be able to:						
1. Apply the mathematical tools to analyze and derive the running time of the algorithms						
2. Demonstrate the major algorithm design paradigms.						
3. Explain major graph algorithms, string matching and geometric algorithms along with their analysis.						
4. Articulating Randomized Algorithms.						
5. Explain the hardness of real-world problems with respect to algorithmic efficiency and learning to cope with it.						
Module:1	Design Paradigms: Greedy, Divide and Conquer Techniques		6 hours			
Overview and Importance of Algorithms - Stages of algorithm development: Describing the problem, Identifying a suitable technique, Design of an algorithm, Derive Time Complexity, Proof of Correctness of the algorithm, Illustration of Design Stages - Greedy techniques: Fractional Knapsack Problem, and Huffman coding - Divide and Conquer: Maximum Subarray, Karatsuba faster integer multiplication algorithm.						
Module:2	Design Paradigms: Dynamic Programming, Backtracking and Branch & Bound Techniques		10 hours			
Dynamic programming: Assembly Line Scheduling, Matrix Chain Multiplication, Longest Common Subsequence, 0-1 Knapsack, TSP- Backtracking: N-Queens problem, Subset Sum, Graph Coloring- Branch & Bound: LIFO-BB and FIFO BB methods: Job Selection problem, 0-1 Knapsack Problem						
Module:3	String Matching Algorithms		5 hours			
Naïve String-matching Algorithms, KMP algorithm, Rabin-Karp Algorithm, Suffix Trees.						
Module:4	Graph Algorithms		6 hours			
All pair shortest path: Bellman Ford Algorithm, Floyd-Warshall Algorithm - Network Flows: Flow Networks, Maximum Flows: Ford-Fulkerson, Edmond-Karp, Push Re-label Algorithm – Application of Max Flow to maximum matching problem						
Module:5	Geometric Algorithms		4 hours			
Line Segments: Properties, Intersection, sweeping lines - Convex Hull finding algorithms: Graham's Scan, Jarvis' March Algorithm.						
Module:6	Randomized algorithms		5 hours			
Randomized quick sort - The hiring problem - Finding the global Minimum Cut.						
Module:7	Classes of Complexity and Approximation Algorithms		7 hours			
The Class P - The Class NP - Reducibility and NP-completeness – SAT (Problem Definition and statement), 3SAT, Independent Set, Clique, Approximation Algorithm – Vertex Cover, Set Cover and Travelling salesman						
Module:8	Contemporary Issues		2 hours			
			Total Lecture hours:			
			45 hours			
Text Book						
1.	Thomas H. Cormen, C.E. Leiserson, R L.Rivest and C. Stein, Introduction to Algorithms, Third edition, MIT Press, 2009.					

Reference Books			
1.	Jon Kleinberg and ÉvaTardos, Algorithm Design, Pearson Education, 1 st Edition, 2014.		
2.	Rajeev Motwani, Prabhakar Raghavan; Randomized Algorithms, Cambridge University Press, 1995 (Online Print – 2013)		
3.	Ravindra K. Ahuja, Thomas L. Magnanti, and James B. Orlin, Network Flows: Theory, Algorithms, and Applications, 1 st Edition, Pearson Education, 2014.		
Mode of Evaluation: CAT, Written assignments, Quiz, FAT.			
Recommended by Board of Studies		04-03-2022	
Approved by Academic Council		No. 65	Date 17-03-2022